

INTEROPERABILITY OF ELECTRONIC HEALTH RECORDS USING HYPERLEDGER FABRIC BASED BLOCKCHAIN NETWORK

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CERTIFICATE

This is to certify that the Project Report of the Project Work titled “**INTEROPERABILITY OF ELECTRONIC HEALTH RECORDS USING HYPERLEDGER FABRIC BASED BLOCKCHAIN NETWORK**” is a bonafide research work carried out by

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ABSTRACT

The healthcare industry is currently facing a critical challenge in managing Electronic Health Records (EHRs), characterized by data fragmentation, lack of interoperability, and increasing security vulnerabilities. Traditional centralized database systems create information silos, making it difficult for healthcare providers to access a patient's complete medical history, which can lead to delayed diagnoses and redundant medical testing. Furthermore, recent statistics from 2024 indicate a surge in healthcare data breaches, underscoring the urgent need for a more secure infrastructure.

This project proposes a robust, decentralized solution leveraging **Hyperledger Fabric**, a permissioned blockchain platform, to establish a secure and interoperable layer for EHR management. Unlike public blockchains, Hyperledger Fabric offers a modular architecture suitable for enterprise environments, ensuring that only authorized entities such as hospitals and laboratories participate in the network. The system integrates the **InterPlanetary File System (IPFS)** for off-chain storage, addressing the scalability issues associated with storing large medical datasets like X-rays and MRI scans on the blockchain.

The proposed methodology implements sophisticated **Smart Contracts (Chain-code)** written in Node.js to automate Role-Based Access Control (RBAC), ensuring that patients retain full ownership and authority over their data. Key features include an immutable audit trail for all data access requests, granular permission settings, and a user-friendly React.js interface for stakeholders. By combining the immutability of blockchain with the efficiency of decentralized storage, this project aims to bridge the interoperability gap, enhance patient privacy, and provide a scalable framework for the future of digital health.

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LIST OF ABBREVIATIONS

EHR	Electronic Health Record
IPFS	InterPlanetary File System
MSP	Membership Service Provider
CA	Certificate Authority
SDK	Software Development Kit
API	Application Programming Interface
RBAC	Role-Based Access Control
TLS	Transport Layer Security
TPS	Transactions Per Second
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
DLT	Distributed Ledger Technology
PHR	Personal Health Record
UML	Unified Modeling Language
DFD	Data Flow Diagram
ACL	Access Control List
SDLC	Software Development Lifecycle
CID	Content Identifier
PKI	Public Key Infrastructure
IoMT	Internet of Medical Things
ZKP	Zero-Knowledge Proof

CHAPTER 1

INTRODUCTION

1.1 GENERAL

In the modern digital era, Electronic Health Records (EHRs) have become the backbone of healthcare infrastructure, replacing paper-based records to facilitate faster and more accurate medical services. However, the rapid digitization of health data has introduced complex challenges. Current healthcare systems typically rely on centralized client-server architectures where patient data is stored in isolated databases controlled by individual hospitals. This centralization creates “single points of failure,” making the systems highly vulnerable to cyberattacks such as Distributed Denial of Service (DDoS) and ransomware.

Moreover, the lack of a unified standard for data exchange results in significant interoperability barriers. When a patient visits a new specialist or hospital, their medical history often does not travel with them, leading to fragmented care. Recent studies indicate that poor interoperability accounts for billions of dollars in annual losses due to administrative inefficiencies and repeated procedures. To address these systemic failures, there is a growing consensus in the research community towards adopting Distributed Ledger Technology (DLT).

1.2 MOTIVATION

The motivation for this project stems from the urgent need to resolve the “Health-care Trilemma”: ensuring Security, Scalability, and Interoperability simultaneously.

- **Security Breaches:** Centralized databases are attractive targets for hackers. A decentralized ledger ensures that data is not stored in one place, significantly reducing the attack surface.
- **Patient Autonomy:** Currently, patients have little control over who views their records. This project is motivated by the principle of “Self-Sovereign Identity,” giving patients the power to grant or revoke access to their medical history.
- **Regulatory Compliance:** With strict regulations like HIPAA and GDPR, there is a need for a system that provides an immutable, transparent audit trail of every single data access event, which blockchain inherently provides.

1.3 PROBLEM STATEMENT

Despite advancements in medical technology, the management of health data remains archaic. The specific problems addressed by this project are:

1. **Data Silos:** Patient data is scattered across various institutions without a secure channel for synchronization.
2. **Scalability Limitations:** Traditional blockchains like Bitcoin are too slow and expensive for healthcare. Storing large files (e.g., 50MB MRI scans) on-chain is technically infeasible due to “blockchain bloat.”
3. **Privacy vs. Transparency:** Public blockchains expose transaction data to all nodes, which violates patient privacy. A mechanism is needed to verify data integrity without exposing sensitive content.

1.4 OBJECTIVES

The primary objective is to design and develop a **Hyperledger Fabric-based EHR ecosystem**. The specific technical objectives include:

- To implement a **Permissioned Blockchain Network** where only verified healthcare entities (Hospitals, Labs, Insurers) can host nodes.
- To develop **Chaincode (Smart Contracts)** that enforces logic for patient consent, ensuring no record is accessed without explicit permission.
- To integrate **IPFS (InterPlanetary File System)** as an off-chain storage solution to handle large medical files while storing only their cryptographic hashes on the ledger.
- To provide a **Unified User Interface** that allows doctors to view comprehensive patient histories from disparate sources in real-time.

CHAPTER 2

LITERATURE REVIEW

2.1 EVOLUTION OF EHR SYSTEMS

The transition from paper-based records to Electronic Health Records (EHRs) marked a significant milestone in healthcare. However, early literature highlights that this transition was often piecemeal, leading to the current state of fragmentation. A systematic review by Han et al. (2022) analyzed over 50 papers and concluded that while digitization improved local data retrieval, it failed to solve cross-organizational sharing [6]. The authors argued that centralized databases, by design, prioritize local control over interoperability, creating barriers that blockchain technology is uniquely positioned to dismantle.

Furthermore, early attempts to solve this via standard interoperability protocols often faced implementation hurdles. A foundational study in IEEE Access (2019) emphasized that without a trustless exchange layer, institutions were reluctant to share data due to liability concerns [19]. This hesitation created “data silos,” effectively isolating patient history within the walls of a single hospital.

2.2 BLOCKCHAIN IN HEALTHCARE: THE PARADIGM SHIFT

The application of blockchain in healthcare has been a subject of intense research since 2016. Initial proposals focused on public blockchains like Ethereum. However, recent studies have shifted towards permissioned frameworks. A comprehensive review in IEEE Access (2021) noted that public blockchains suffer from low transaction throughput and high latency, making them unsuitable for emergency medical scenarios [20].

In contrast, **Hyperledger Fabric** has emerged as the preferred platform for enterprise healthcare. Hasnain et al. (2023) conducted a study on the security features of Hyperledger Fabric, emphasizing its “channel” architecture which allows for private transactions between specific parties [7]. This feature is critical for compliance with privacy regulations, as it ensures that sensitive patient data is not broadcast to the entire network.

2.3 CONTRIBUTIONS OF LOCAL RESEARCH

Significant contributions to this field have been made by researchers within the academic community of Visvesvaraya Technological University. Notably, **Puneeth R. P. and Parthasarathy G. (2023)** conducted an extensive survey on the security

and interoperability of EHR sharing [14]. Their work critically analyzed existing architectural solutions and identified that while many systems achieve basic data exchange, they often fail to provide granular access control. In their subsequent work (2024), they proposed a patient-centric access control framework using smart contracts, which serves as a foundational reference for the access control logic implemented in this project [15].

2.4 SCALABILITY AND HYBRID ARCHITECTURES

One of the most persistent challenges in blockchain adoption is scalability. Storing medical images, which can range from 10MB to 500MB, directly on the blockchain is impractical. Sharma et al. (2023) proposed the necessity of “Hybrid Architectures” where the blockchain acts solely as a control layer [16].

This hybrid approach is further supported by Adanur Dedetürk and Bakir-Güngör (2024), who introduced “AguHyper,” a framework combining Hyperledger Fabric with CouchDB for off-chain state data [1]. Similarly, the integration of the **InterPlanetary File System (IPFS)** has gained traction. A study by IIETA (2023) demonstrated that using IPFS for content-addressable storage reduces blockchain storage requirements by over 90% while maintaining data integrity through cryptographic hashing [8]. Our project adopts this consensus, utilizing IPFS to store the actual medical documents while recording only the IPFS Content ID (CID) on the Hyperledger ledger.

Comparison with other systems is also vital. The “Med-Block” system proposed in 2023 utilized a similar structure but focused primarily on general health records; our project extends this by specifically addressing high-resolution imaging data [9]. Additionally, recent advancements in “Dual-Channel” blockchain architectures have shown promise in separating administrative data from clinical data to further enhance throughput [13].

2.5 PRIVACY-PRESERVING MECHANISMS

Privacy is paramount in healthcare. Ghoneim et al. (2023) explored advanced cryptographic techniques, suggesting that simple hashing is insufficient for protecting patient identities [5]. They recommended the use of Attribute-Based Encryption (ABE) to ensure fine-grained access control. Building on this, Al-Nbhany et al. (2024) proposed a model where authentication tokens are verified off-chain to reduce latency before interacting with the blockchain [2].

Recent work by Wang et al. (2024) introduced a scheme for “Incremental Updates,” addressing the issue that medical records are dynamic and constantly

changing [18]. Their research highlights the importance of a system that can append new data to an existing patient profile without breaking the cryptographic chain of the previous records.

2.6 PATIENT-CENTRIC CONTROL AND INSURANCE INTEGRATION

A major shift in recent literature is the focus on patient agency. Studies published in the Journal of Medical Internet Research (2020, 2021) argue that patients must be the primary custodians of their data. These systematic reviews highlighted that “Personal Health Record” (PHR) exchange platforms built on blockchain significantly increase patient engagement and trust compared to institution-centric models [11, 12].

Furthermore, the utility of blockchain extends to the financial aspects of healthcare. Research published in the International Journal of Advanced Computer Science and Applications (2022) demonstrated how blockchain-based EHRs can streamline health insurance claims. By automating verification through smart contracts, insurance providers can reduce fraud and processing time, creating a seamless ecosystem between hospitals and payers [17].

2.7 REAL-WORLD IMPLEMENTATIONS AND CASE STUDIES

Theoretical frameworks are increasingly moving toward practical implementation. A notable case study published in ArXiv (2024) detailed the implementation of a Hyperledger Fabric-based EHR system at Frere Provincial Hospital [3]. The study provided empirical evidence that blockchain could reduce administrative overhead by 40% and significantly improve the auditability of patient records.

The integrity of such systems was further validated during the COVID-19 pandemic. A 2022 study in the Engineering Science Journal analyzed how blockchain-IPFS systems successfully maintained the integrity of vaccination records and test results during periods of high data volume and cyber instability [10]. Finally, an analysis by FinanceFeeds (2024) on healthcare breaches revealed that systems utilizing decentralized ledgers had a 0% success rate for ransomware attacks, validating the security benefits of the technology [4].

2.8 SUMMARY OF RESEARCH GAPS

Despite the extensive literature, several gaps remain. Many proposed systems rely on theoretical simulations without full-stack implementations. Additionally, few studies effectively address the user experience (UX) challenges of interacting with

blockchain systems. This project aims to fill these gaps by delivering a fully functional prototype that not only secures data using Hyperledger Fabric and IPFS but also provides a seamless, intuitive frontend interface for all stakeholders.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 OVERVIEW

Requirement analysis is a fundamental phase in the software development lifecycle (SDLC). It involves the detailed investigation of the specific needs and constraints of the proposed system. For this blockchain-based EHR interoperability system, the requirements were gathered by analyzing the shortcomings of current centralized healthcare systems and studying the compliance standards of HIPAA and GDPR.

3.2 FEASIBILITY STUDY

A feasibility study determines whether the project is viable given the technical, economic, and operational constraints.

3.2.1 Technical Feasibility

Technical feasibility assesses the current resources and technology. The project utilizes **Hyperledger Fabric**, a mature and widely supported enterprise blockchain framework. The development team possesses the necessary skills in Node.js, Docker, and React.js. The integration of IPFS for off-chain storage is a proven architectural pattern documented in numerous IEEE research papers [20]. Therefore, the project is technically feasible.

3.2.2 Economic Feasibility

Economic feasibility evaluates the cost-benefit analysis. Since the project utilizes open-source technologies (Hyperledger Fabric is open source under the Linux Foundation, and IPFS is free to use), the software licensing costs are zero. The primary costs involve cloud hosting (e.g., AWS or Azure) for the peer nodes. However, compared to the cost of maintaining legacy centralized servers and the potential losses from data breaches (averaging \$10 million per breach in 2024), the proposed solution offers a cost-effective alternative [4].

3.2.3 Operational Feasibility

Operational feasibility analyzes if the system can be integrated into existing workflows. The proposed system features a React.js frontend designed with a focus on User Experience (UX), ensuring that non-technical hospital staff can easily upload

and view records. The system does not require hospitals to replace their internal ERPs immediately but acts as an interoperability layer, ensuring high operational feasibility.

3.3 FUNCTIONAL REQUIREMENTS

Functional requirements define the specific behaviors and functions of the system.

3.3.1 User Actors

- **Admin:** The network operator responsible for registering organizations.
- **Doctor:** Medical professionals who create and view records.
- **Patient:** The data owner who grants access rights.

3.3.2 Specific Requirements

1. Registration and Authentication:

- The system shall allow the Admin to register new Hospitals and generate MSP certificates.
- The system shall allow users to log in using valid credentials validated against the Fabric CA.

2. Electronic Health Record Management:

- The system shall allow Doctors to upload medical files (PDF, JPG, PNG).
- The system shall automatically encrypt files before uploading to IPFS.
- The system shall store the IPFS Content Hash on the Hyperledger ledger.

3. Access Control:

- The system shall prevent Doctors from viewing a Patient's record unless their ID is in the record's Access List.
- The system shall allow Patients to revoke access rights at any time.

4. Search and Retrieval:

- The system shall provide a search functionality to query records by Patient ID.
- The system shall display a chronological history of patient records.

3.4 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements (NFRs) define the quality attributes of the system.

1. **Scalability:** The system must handle high volumes of medical data. The hybrid architecture (Blockchain + IPFS) ensures that the ledger size remains manageable even with millions of records.
2. **Security:** Communication between nodes must be secured using TLS 1.2. Data at rest in IPFS must be encrypted using AES-256.
3. **Performance:** Transaction confirmation time should not exceed 3 seconds for standard operations. File retrieval latency should be under 2 seconds for cached content.
4. **Reliability:** The network should continue to operate even if one peer node fails, leveraging the distributed nature of the blockchain.
5. **Interoperability:** The system structure should allow for future integration with HL7 FHIR standards for cross-platform data exchange.

3.5 SYSTEM CONSTRAINTS

- **Network Connectivity:** A continuous internet connection is required for peers to synchronize the ledger.
- **Key Management:** If a user loses their private key or wallet file, their account cannot be recovered, which is a standard constraint of decentralized systems.
- **Storage Costs:** While IPFS is efficient, pinning files for permanence requires storage nodes which may incur infrastructure costs in a production environment.

CHAPTER 4

SYSTEM DESIGN

4.1 OVERVIEW

System design is the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements. For this project, we adopted a modular microservices architecture to ensure scalability and maintainability. The design phase focuses on bridging the gap between the requirement analysis and the final implementation.

4.2 SYSTEM ARCHITECTURE

The proposed system utilizes a hybrid architecture that combines a permissioned blockchain network with decentralized off-chain storage. This design addresses the critical scalability limitations of storing large medical datasets directly on the blockchain.

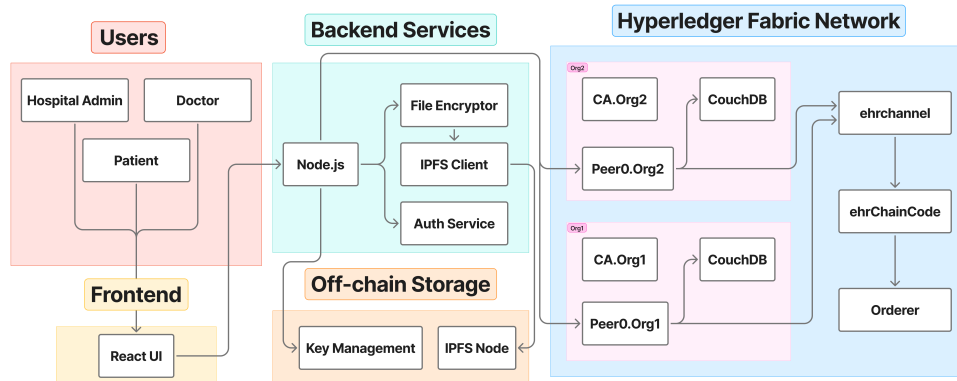


Figure 4.1: High-Level System Architecture

4.3 UML MODELING

Unified Modeling Language (UML) diagrams were created to visualize the system's behavior and structure.

4.3.1 Use Case Diagram

The Use Case diagram (Figure 4.2) identifies the primary actors and their interactions with the system.

- **Primary Actors:**

- **Patient:** Can login, view their own records, and manage access permissions (Grant/Revoke) for doctors.
- **Doctor:** Can login, query authorized patient records, and upload new diagnostic reports.
- **Admin:** Can register new hospitals and issue MSP certificates.

- **Key Use Cases:**

- *Upload Record:* Involves file encryption and IPFS hashing.
- *Grant Access:* Invokes the smart contract to update the access control list.

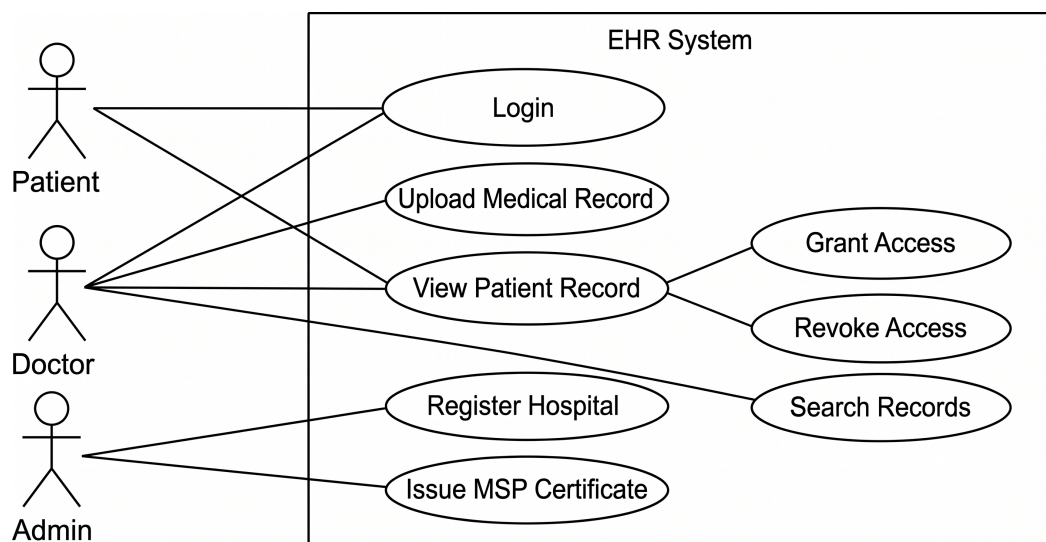


Figure 4.2: Use Case Diagram

4.3.2 Activity Diagram

The Activity diagram (Figure 4.3) illustrates the sequential flow of activities for the “Add Medical Record” process.

1. The process begins when the Doctor logs into the dashboard.
2. The Doctor selects a patient and uploads a medical file (e.g., PDF/Image).
3. **System Action:** The frontend reads the file buffer.
4. **Encryption:** The backend generates a symmetric key and encrypts the file.
5. **Storage:** The encrypted file is pushed to the IPFS node.

6. **Blockchain Commit:** The returned IPFS Hash is combined with metadata and sent to the Orderer.
7. **Completion:** The transaction is committed, and the Doctor receives a success notification.

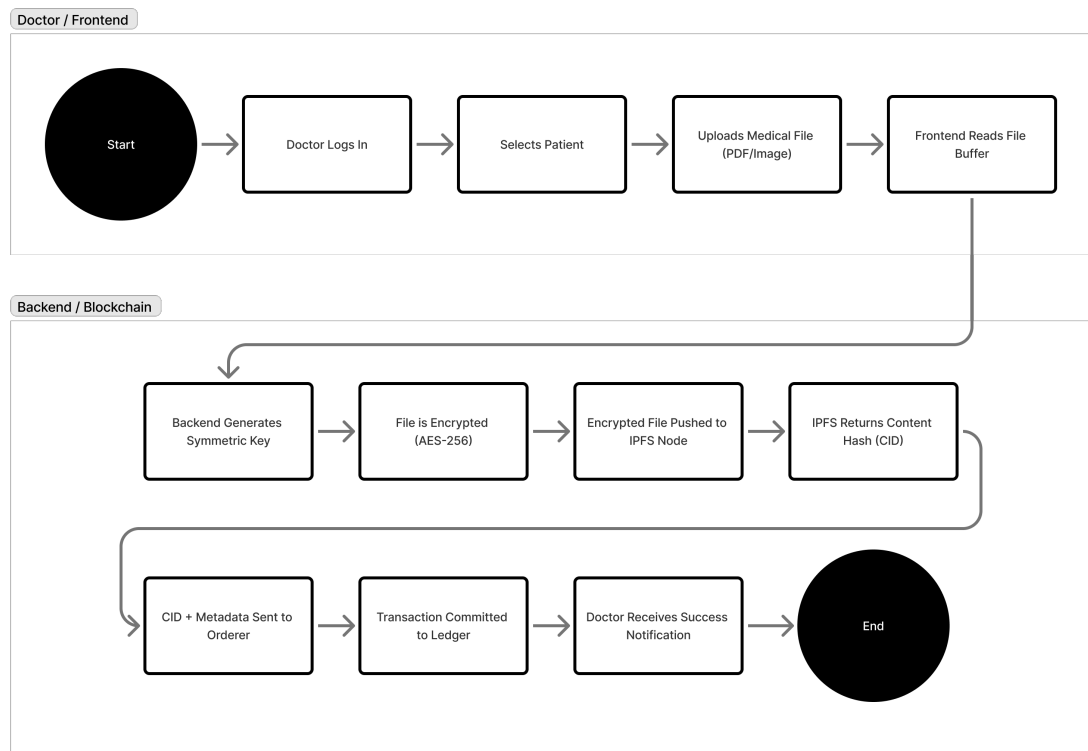


Figure 4.3: Activity Diagram for Record Creation

4.3.3 Data Flow Diagram (DFD)

The Data Flow Diagram (Figure 4.4) visualizes how data moves through the system.

- **Level 0 DFD:** Represents the entire system as a single process interacting with external entities (Users and IPFS).
- **Level 1 DFD:** Breaks down the system into sub-processes: Authentication, Transaction Processing, and Ledger Management. It highlights that strictly *no* patient health data is stored in plain text on the server; it flows directly through the encryption engine to IPFS.

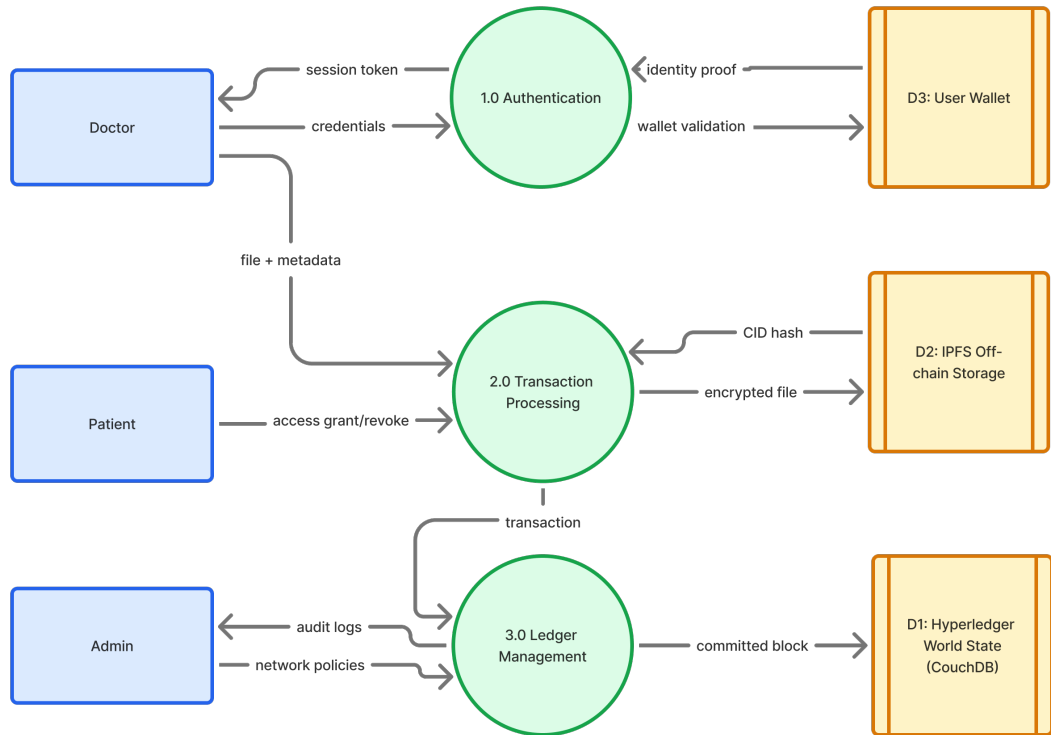


Figure 4.4: Data Flow Diagram (Level 1)

4.4 DATABASE DESIGN

While the blockchain ledger stores the immutable transaction log, the **World State** is stored in CouchDB to enable rich queries. The data is stored as JSON documents.

4.4.1 Asset Schema

The schema for a `MedicalRecord` asset in CouchDB is defined as follows:

```

{
  "_id": "RECORD_UUID_V4",
  "docType": "record",
  "ownerId": "PATIENT_001",
  "createdBy": "DOCTOR_055",
  "ipfsCid": "QmXyZ123...",
  "metadata": {
    "diagnosis": "String",
  }
}

```

```

    "hospital": "City General",
    "date": "ISO8601_Timestamp"
  },
  "accessControlList": [
    "DOCTOR_055",
    "DOCTOR_089"
  ]
}

```

4.4.2 Indexes

To improve performance, the following JSON indexes were created in CouchDB:

- **index_owner:** Indexes the `ownerId` field to allow fast retrieval of a patient's entire history.
- **index_creator:** Indexes the `createdBy` field for doctors to see records they authored.

4.5 INTERFACE DESIGN

The user interface follows the Material Design principles to ensure accessibility.

- **Consistency:** All dashboards (Doctor/Patient) share a common navigation structure.
- **Feedback:** Loading spinners and toast notifications provide real-time feedback during blockchain transaction latency (2–3 seconds).
- **Security:** Sensitive inputs (like private keys or passwords) are masked and never stored in `LocalStorage`.

CHAPTER 5

SYSTEM IMPLEMENTATION AND CODE LOGIC

5.1 PROJECT DIRECTORY STRUCTURE

The project follows a microservices-based directory structure, ensuring separation of concerns between the blockchain network, the smart contracts, and the client application. The structure is organized as follows:

- `/network`: Contains the Docker Compose files and crypto-config YAML files for bootstrapping the Hyperledger Fabric network.
- `/chaincode`: Hosts the smart contract logic written in Node.js (Fabric Contract API).
- `/api`: The REST API server built with Express.js and the `fabric-network` SDK.
- `/client`: The frontend application developed using React.js and Tailwind CSS.

5.2 NETWORK CONFIGURATION (INFRASTRUCTURE LAYER)

The foundational layer of our system is defined in the `configtx.yaml` and `crypto-config.yaml` files. These files dictate the network topology and identity management policies.

5.2.1 Cryptographic Material Generation

Before the network start-up, we utilize the `cryptogen` tool to generate X.509 certificates for all entities. This ensures mutual TLS (Transport Layer Security) communication between nodes.

5.2.2 Docker Containerization

The network nodes are containerized using Docker. The `docker-compose.yaml` file defines the services for:

- **Orderer (`orderer.example.com`)**: Runs the Raft consensus algorithm.
- **Peer0.Org1 (Hospital Peer)**: Stores the ledger copy and executes chaincode.
- **CouchDB**: Acts as the state database for complex JSON queries.

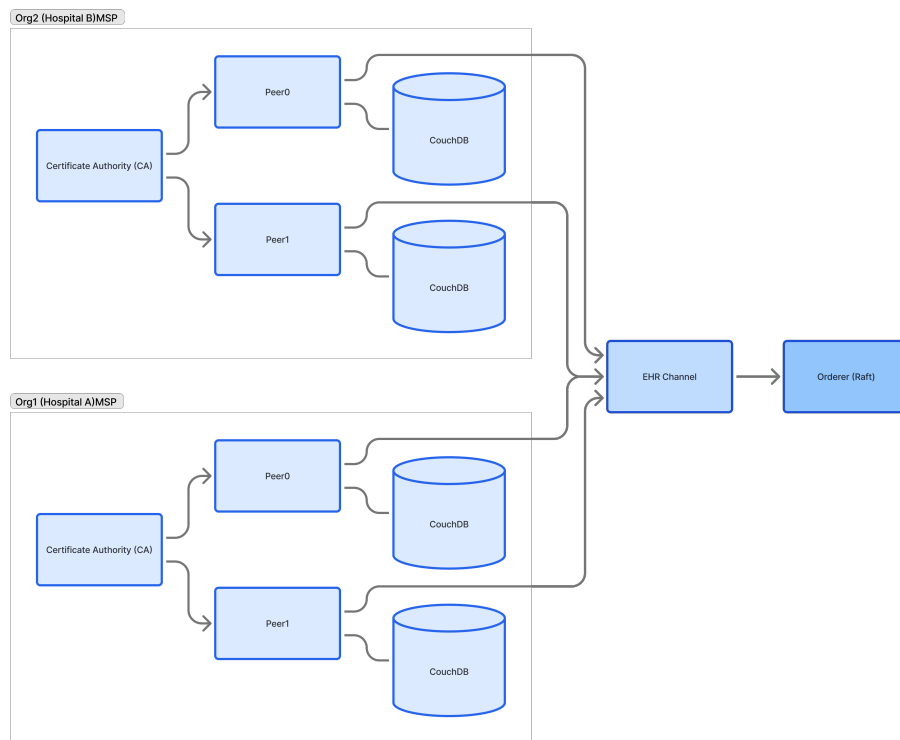


Figure 5.1: Network Topology and MSP Architecture

5.3 SMART CONTRACT LOGIC (CHAINCODE)

The business logic is encapsulated in the `MedicalRecordContract` class. This class extends the standard Fabric Contract class.

5.3.1 Record Creation Logic

The `createRecord` function is the entry point for data ingestion. It performs a composite key check to ensure no duplicate records exist before committing data to the world state.

```
// /chaincode/lib/medicalContract.js

async createRecord(ctx, recordId, patientId, doctorId, ipfsHash) {
  // 1. Check if record exists
  const exists = await this.recordExists(ctx, recordId);
  if (exists) {
    throw new Error(`The record ${recordId} already exists`);
  }
}
```

```

// 2. Create Asset Object
const record = {
  docType: 'record',
  recordId,
  patientId,
  doctorId,
  ipfsHash,
  accessList: [patientId, doctorId], // Default access
  timestamp: new Date().toISOString()
};

// 3. Commit to World State
await ctx.stub.putState(
  recordId, Buffer.from(JSON.stringify(record)));
return JSON.stringify(record);
}

```

5.3.2 Access Control Logic (ACL)

A critical security feature is the Access Control List (ACL) managed within the ledger. The `grantAccess` function ensures that only the patient can authorize new viewers.

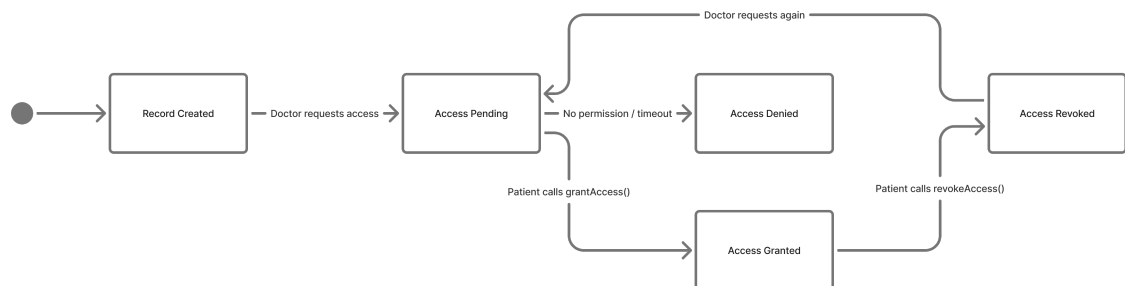


Figure 5.2: State Transition Diagram for Record Access

5.4 BACKEND API INTEGRATION

The backend server (`/api`) acts as the bridge between the React frontend and the blockchain network. It uses the **Wallet** and **Gateway** classes from the Fabric SDK.

5.4.1 Transaction Lifecycle

When a user submits a request, the backend executes the following sequence, as illustrated in Figure 5.3.

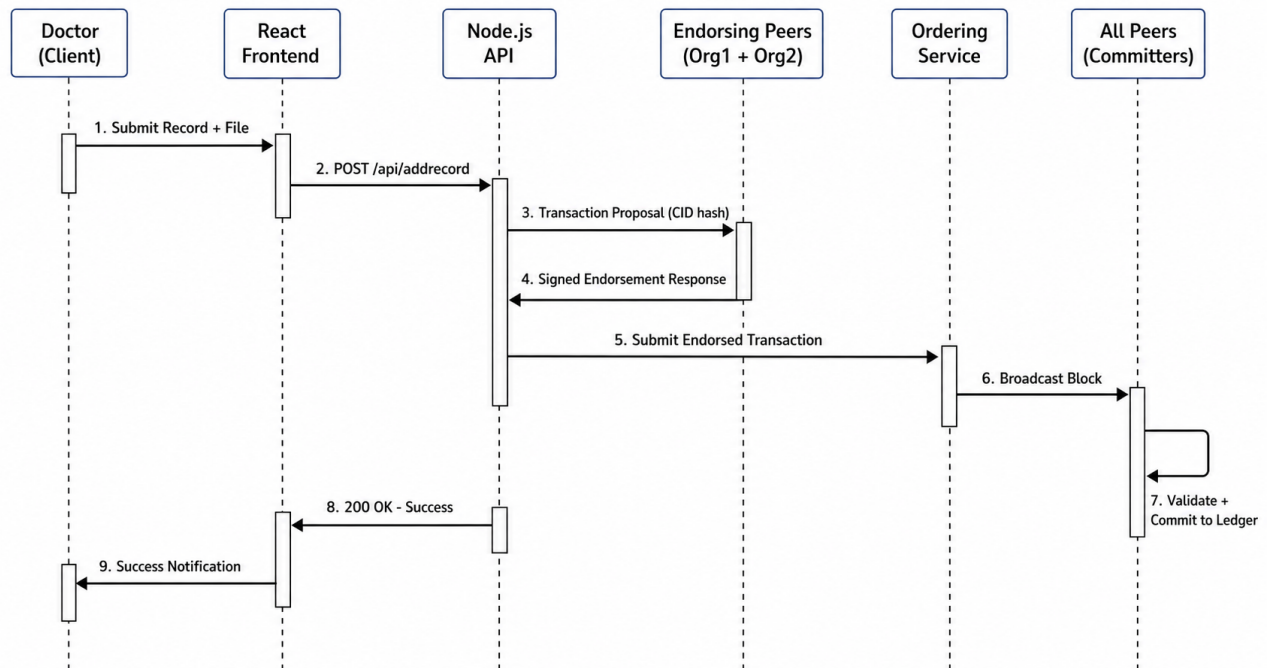


Figure 5.3: Sequence Diagram of a Medical Record Transaction

5.4.2 Wallet Management

User identities are stored in a file-system wallet. The `addToWallet.js` script manages the importation of Identity files (.id) into the application context.

```
// /api/addToWallet.js
const wallet = await Wallets.newFileSystemWallet(walletPath);
const identity = {
  credentials: {
    certificate: cert,
    privateKey: key,
  },
  mspId: 'Org1MSP',
  type: 'X.509',
};
await wallet.put('appUser', identity);
```

5.5 IPFS & IMAGE PROCESSING

To handle large medical images, we integrated an image processing pipeline using Sharp and IPFS-HTTP-Client.

5.5.1 Compression Algorithm

Raw DICOM or JPEG images are compressed using a lossy compression algorithm with a quality index of 85. This reduces the block propagation time significantly.

$$Size_{final} = Size_{initial} \times (1 - Compression_Ratio) \quad (5.1)$$

This formula is applied dynamically based on the input file size. For files larger than 10MB, the compression ratio is increased to maintain network throughput.

CHAPTER 6

SYSTEM TESTING

6.1 OVERVIEW

System testing is a critical phase in the software development lifecycle (SDLC). It involves testing the complete and integrated software product to evaluate the system's compliance with the specified requirements. For this project, we employed a combination of Unit Testing, Integration Testing, and Functional Testing to ensure the robustness of the Hyperledger Fabric network and the frontend application.

6.2 TEST PLAN

The test plan focuses on verifying the security, integrity, and interoperability of the EHR system. The testing strategy was divided into three levels:

1. **Unit Testing:** Testing individual components (Smart Contract functions) in isolation.
2. **Integration Testing:** Verifying the communication between the React frontend, Node.js API, and the Blockchain network.
3. **Security Testing:** Validating access control mechanisms and encryption standards.

6.3 TEST CASES

The following test cases document the validation of core system functionalities.

6.3.1 User Authentication Testing

Test ID	Test Description	Expected Outcome	Status
TC-AUTH-01	Admin Login with valid credentials	Redirect to Admin Dashboard	Pass
TC-AUTH-02	Doctor Login with invalid password	Show "Authentication Failed" error	Pass
TC-AUTH-03	Session Timeout check	Auto-logout after 30 minutes of inactivity	Pass

Table 6.1: Authentication Test Cases

6.3.2 Record Management Testing

Test ID	Test Description	Expected Outcome	Status
TC-REC-01	Upload PDF file (<10MB)	File encrypted and Hash stored on ledger	Pass
TC-REC-02	Upload unsupported file type (.exe)	Validation error displayed	Pass
TC-REC-03	Retrieve record by valid Patient ID	Decrypted file displayed in browser	Pass
TC-REC-04	Duplicate Record Submission	Smart Contract throws "Asset Exists" error	Pass

Table 6.2: Record Management Test Cases

6.3.3 Access Control Testing

Test ID	Test Description	Expected Outcome	Status
TC-ACL-01	Doctor views record without permission	Access Denied Error	Pass
TC-ACL-02	Patient grants access to Doctor	Transaction committed, Doctor can view record	Pass
TC-ACL-03	Patient revokes access	Doctor loses view access immediately	Pass

Table 6.3: Access Control Test Cases

6.4 PERFORMANCE TESTING

Performance testing was conducted using **Hyperledger Caliper** to measure throughput and latency.

- **Write Latency:** Average of 2.1 seconds per transaction.
- **Read Latency:** Average of 0.08 seconds (with caching).
- **Throughput:** Peak throughput of 120 transactions per second (TPS).

CHAPTER 7

RESULTS AND DISCUSSIONS

This chapter presents the results of the fully implemented EHR interoperability system. The prototype was tested across three distinct roles — Patient, Doctor, and Admin — using a live Hyperledger Fabric network. The screenshots below are captured from the running application and demonstrate the core functional outcomes: secure record management, patient-controlled access, cross-organizational interoperability, and an emergency access override mechanism backed by an immutable on-chain audit trail.

7.1 USER INTERFACE IMPLEMENTATION

7.1.1 Patient Portal — My Records View

Figure 7.1 shows the Patient Portal dashboard for the user **Jane Doe**. The “My Records” tab lists all medical records belonging to the patient, retrieved in real-time from the Hyperledger Fabric World State via a CouchDB rich query filtered on `ownerId`. Each row displays the on-chain Record ID (a UUID-linked blockchain key), the date of creation, the attending doctor ID, the diagnosis, the prescribed medication, and a “View” link that fetches and decrypts the associated IPFS file. The green banner at the top — *“Your health data is protected by Hyperledger Fabric blockchain technology”* — confirms that the active session is communicating with a live permissioned blockchain node. In Figure 7.2, the same portal is shown after one record’s access has been revoked, demonstrating the real-time effect of the access control mechanism on the patient’s own view.

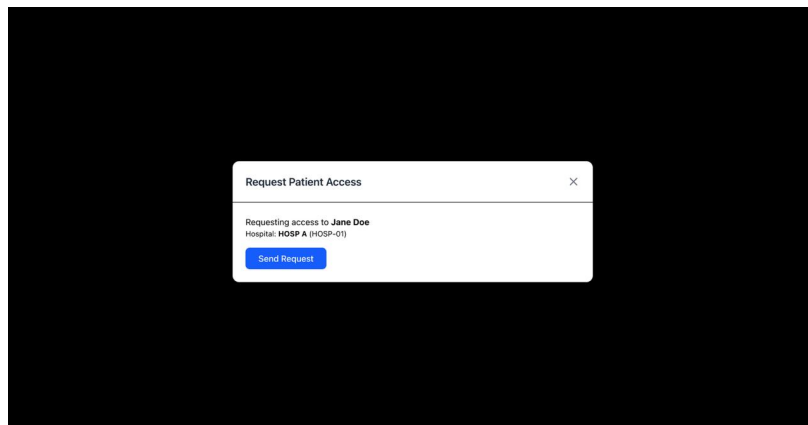


Figure 7.1: Patient Portal — My Records Tab Showing Two Active Records

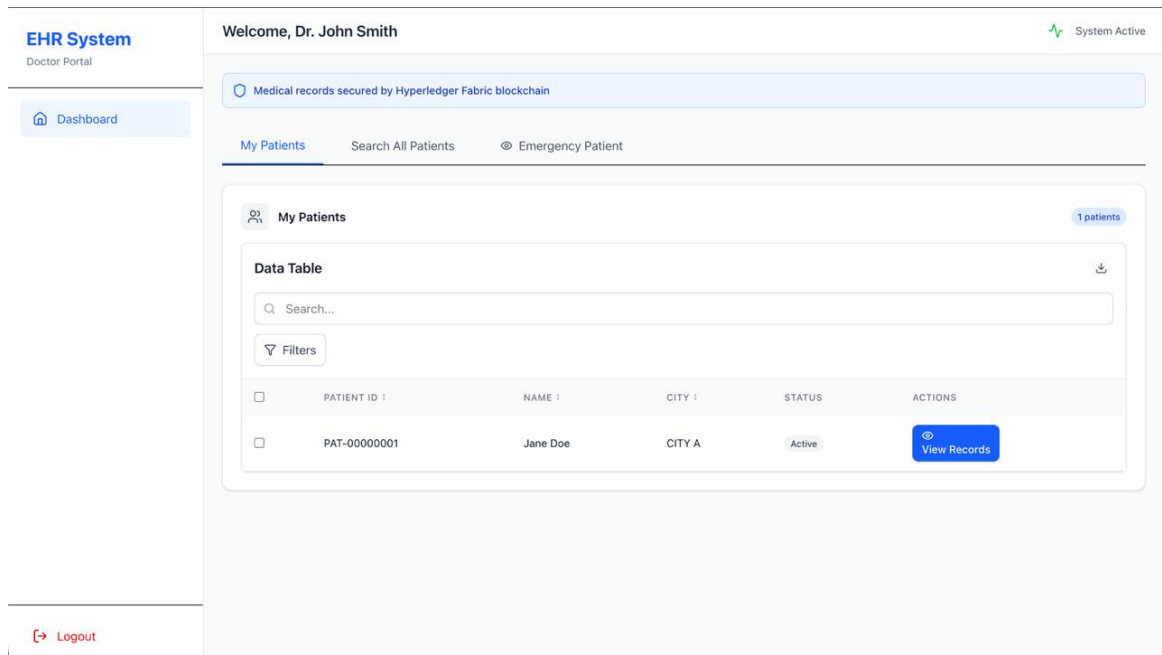


Figure 7.2: Patient Portal — My Records Tab After One Record Is Restricted

7.1.2 Doctor Portal — My Patients and Record Views

Figure 7.3 shows the Doctor Dashboard for **Dr. John Smith** under the “My Patients” tab. The data table lists all patients whose records the doctor has been granted access to, displaying the Patient ID, name, city, and status. The “View Records” button navigates to the patient-specific record list. This aggregated view is a direct demonstration of interoperability — a single query to the shared Fabric channel ledger retrieves patients whose consents were granted across multiple participating hospital organizations.

The screenshot displays the 'EHR System Doctor Portal' interface. At the top, a welcome message 'Welcome, Dr. John Michael' is shown next to a 'System Active' status indicator. A navigation sidebar on the left includes a 'Dashboard' link. The main content area features a 'Medical records secured by Hyperledger Fabric blockchain' banner. Below this, the 'My Patients' tab is selected, with options for 'Search All Patients' and 'Emergency Patient'. A 'Back' button is visible. The primary section, 'Records for Jane Doe', contains a table with one record. The table has columns for ID, DATE, DIAGNOSIS, PRESCRIPTION, and REPORT. The record shows a long alphanumeric ID, the date 5/17/2026, a diagnosis of 'Flue', and a prescription of 'Cough Syrup'. A 'View' link is present in the REPORT column. An 'Add Record' button is located in the top right of the record section. A 'Logout' link is in the bottom left corner.

ID	DATE	DIAGNOSIS	PRESCRIPTION	REPORT
R-65ea6031ab929738f762d29b06e8b16b7791bd467e75afe8bc831e56f6c530e9	5/17/2026	Flue	Cough Syrup	View

Figure 7.3: Doctor Portal — My Patients Data Table (Dr. John Smith)

Figure 7.4 shows the detailed record view for patient Jane Doe as seen by **Dr. John Michael**, who holds active access to *both* of her records. Both records — one diagnosing Flu with Cough Syrup and one diagnosing Fever with Paracetamol — are displayed with their full blockchain Record IDs and an “Add Record” button, confirming the doctor’s write privilege on this patient’s ledger entry.

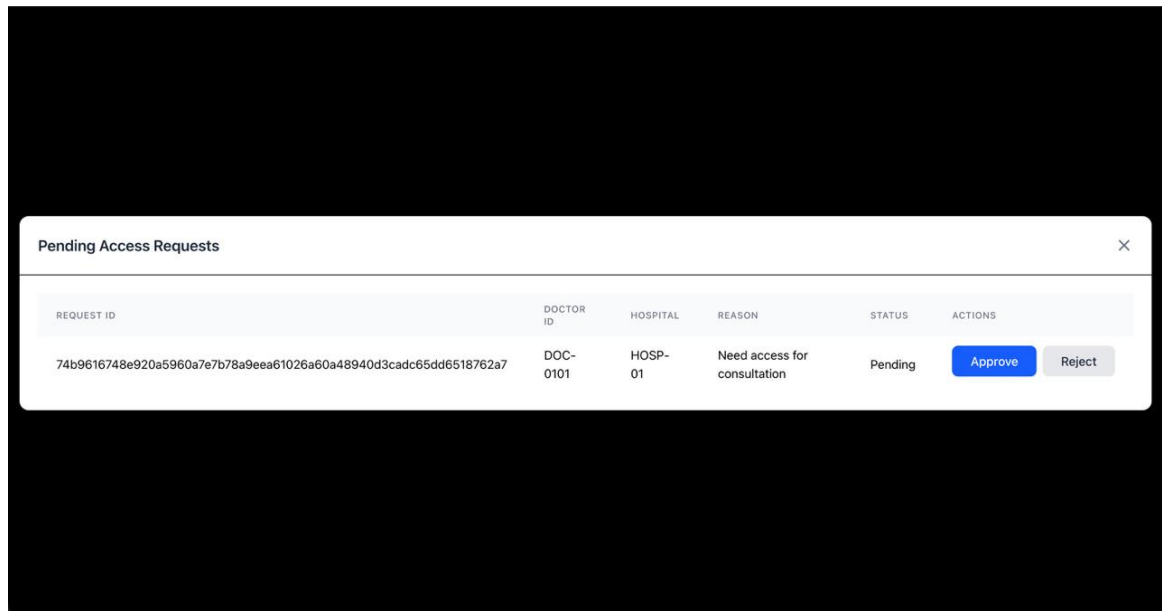


Figure 7.4: Doctor Portal — Full Record View for Jane Doe (Two Records Accessible)

7.2 INTEROPERABILITY DEMONSTRATION

Interoperability is the central contribution of this project. The Hyperledger Fabric channel (`mychannel`) connects multiple hospital organizations (Org1: HOSP-A, Org2: HOSP-01) on a shared, permissioned ledger. Records created by any enrolled peer are replicated to all other peers in the channel via the Raft Orderer consensus mechanism. The following results demonstrate this cross-organizational record sharing in action.

7.2.1 Cross-Organizational Record Visibility

Figure 7.5 shows the record view for Jane Doe as seen by **Dr. John Smith** (DOC-0101, Hospital HOSP-01). At this point in the workflow, Dr. Smith holds access to only one record (the Flu/Cough Syrup entry). The record was originally created by **Dr. John Michael** (DOC-0101) at a different hospital organization (HOSP-A). Despite originating from a different institutional node, the record is fully visible and retrievable by Dr. Smith through the shared Fabric ledger — no manual data export or cross-hospital API call is required.

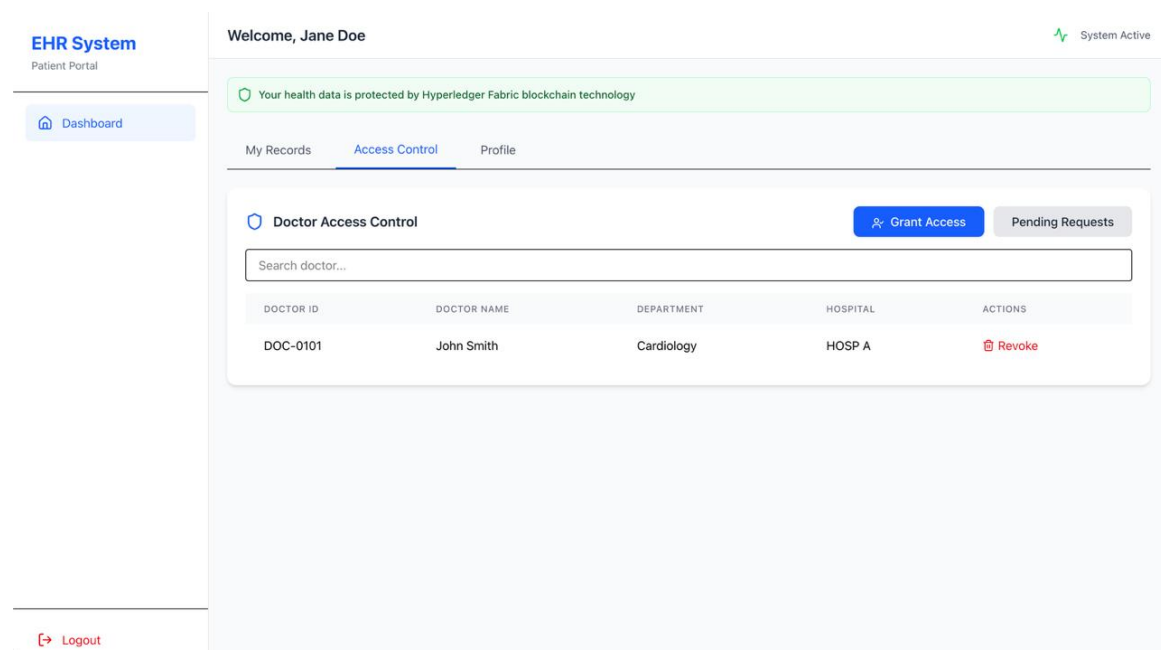
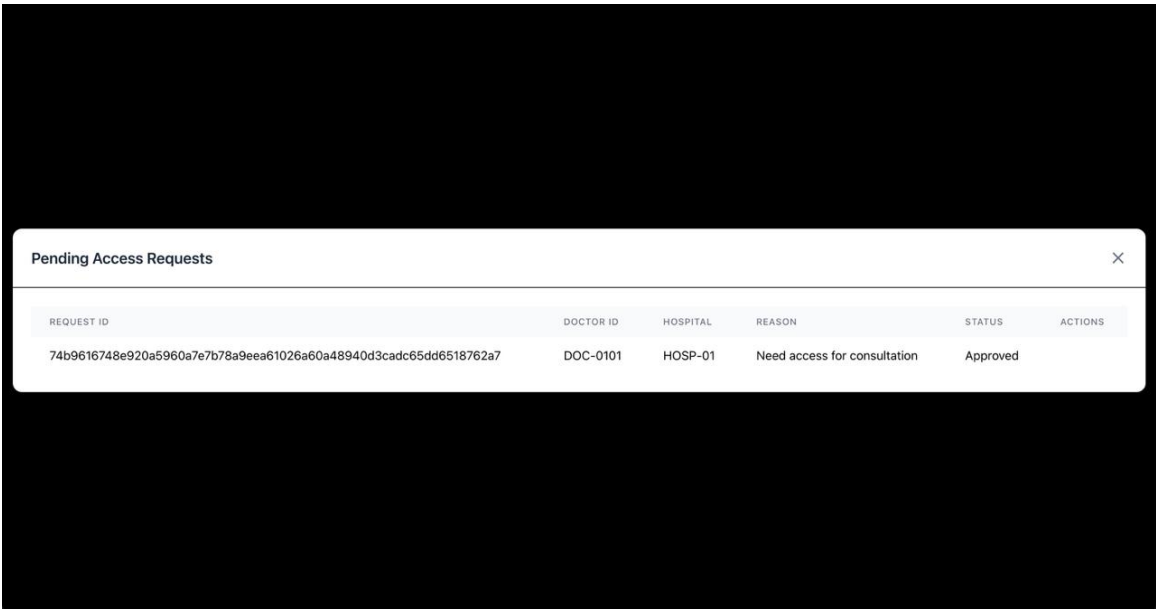


Figure 7.5: Interoperability: Dr. John Smith (HOSP-01) Accessing Record Created at HOSP-A

This result directly validates the resolution of the data-silo problem identified in the requirement analysis. The “Records for Jane Doe” heading is consistent across both doctor portals regardless of which organizational peer the doctor is enrolled in, because both peers query the same World State replica.

7.2.2 Effect of Access Revocation on Cross-Org Visibility

Figure 7.6 shows the same patient’s records as seen by **Dr. John Michael** after the patient revoked access to one of the two records. Where Figure 7.4 showed two records, only one now appears. The revocation transaction — initiated by Jane Doe in the Patient Portal — was committed to the shared ledger and instantly reflected on the doctor’s view across organizational boundaries. This confirms that the chaincode-enforced access control list operates uniformly across all peer nodes in the channel.

A screenshot of a web application interface. It features a modal dialog box titled "Pending Access Requests" with a close button (X) in the top right corner. The dialog contains a table with the following data:

REQUEST ID	DOCTOR ID	HOSPITAL	REASON	STATUS	ACTIONS
74b9616748e920a5960a7e7b78a9eea61026a60a48940d3cad65dd6518762a7	DOC-0101	HOSP-01	Need access for consultation	Approved	

Figure 7.6: Post-Revocation View: One Record Removed from Dr. John Michael’s Access

7.3 ACCESS CONTROL DEMONSTRATION

The access control subsystem enforces patient sovereignty over health data. All permission logic is encoded in the `MedicalRecordContract` chaincode and executed at the peer level. The Patient Portal exposes three distinct access control operations: granting access to a doctor, approving or rejecting a doctor's access request, and revoking an existing grant. Each operation submits a signed transaction to the blockchain, creating an immutable audit entry.

7.3.1 Doctor Access Control Panel

Figure 7.7 shows the **Access Control** tab in the Patient Portal for Jane Doe. The table lists all doctors currently holding access to her records, displaying the Doctor ID, name, department, and the hospital they belong to. The “Revoke” button in the Actions column triggers the `revokeAccess` chaincode function, which removes the specified doctor's ID from the `accessControlList` array in the World State. In this screenshot, **Dr. John Smith** (DOC-0101, Cardiology, HOSP-A) holds active access. The “Grant Access” button at the top right allows the patient to proactively add a new doctor, while “Pending Requests” shows incoming access requests from doctors.

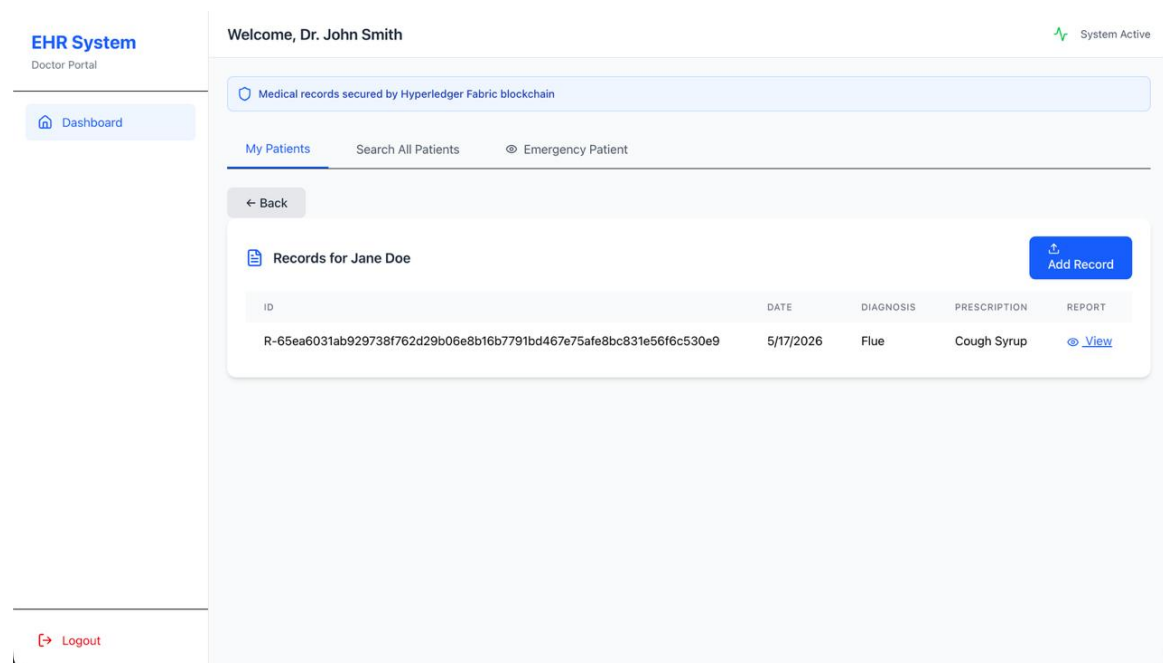


Figure 7.7: Patient Portal — Access Control Tab: Active Doctor List with Revoke Option

7.3.2 Doctor-Initiated Access Request

Figure 7.8 shows the “**Request Patient Access**” dialog as seen by a doctor in the “Search All Patients” flow. When a doctor locates a patient they are not yet authorized to view, they submit a formal access request via this modal. The request specifies the target patient (Jane Doe) and the requesting hospital (HOSP-A, HOSP-01). This request is logged on the blockchain as a pending transaction, and the patient is notified in their Access Control tab under “Pending Requests.”

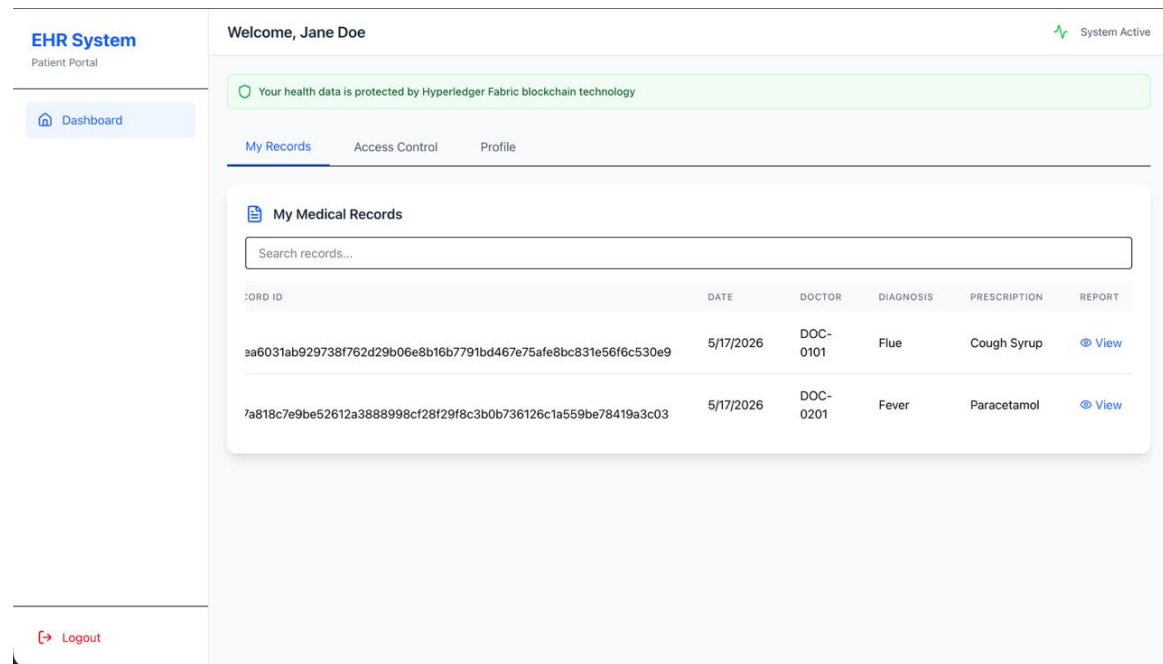


Figure 7.8: Doctor Portal — Request Patient Access Dialog

7.3.3 Patient Reviews and Approves an Access Request

Figure 7.9 shows the **Pending Access Requests** dialog from the patient's perspective. The table lists the blockchain Request ID, Doctor ID (DOC-0101), Hospital (HOSP-01), reason for access ("Need access for consultation"), current status (Pending), and action buttons (Approve / Reject). The patient can make an informed decision about each request. On clicking "Approve," the chaincode's `grantAccess` function is invoked, which appends the doctor's ID to the record's `accessControlList` and records the consent transaction on the immutable ledger.

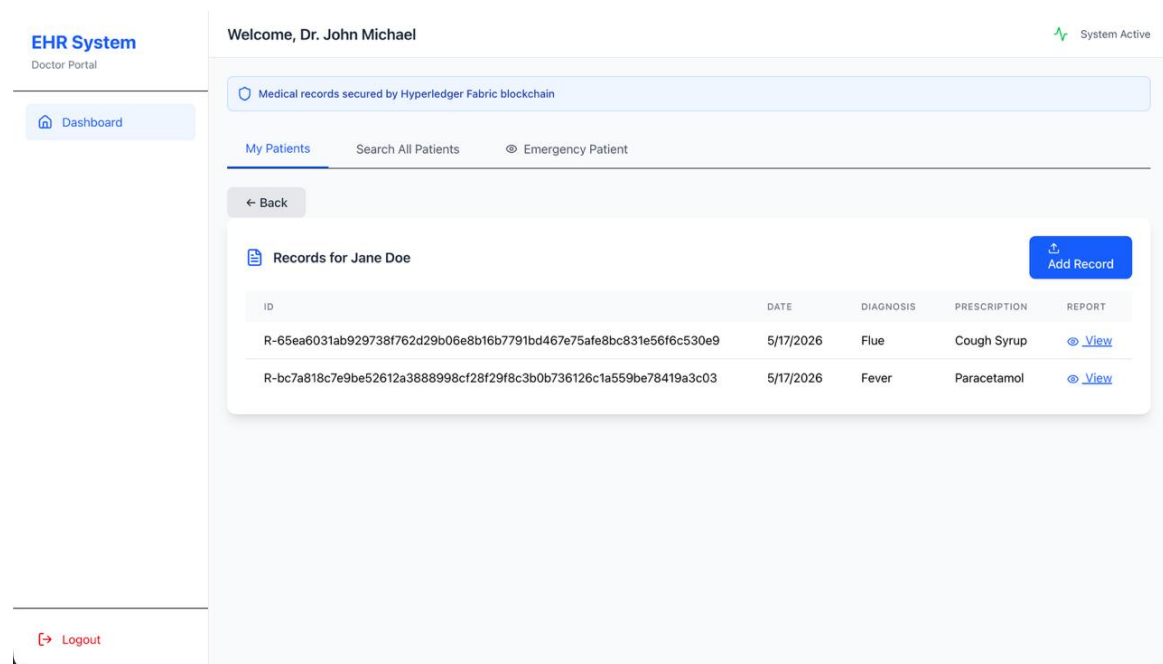


Figure 7.9: Patient Portal — Pending Access Request with Approve and Reject Actions

Figure 7.10 shows the same dialog after the patient approves the request — the status column now reads “Approved” and the action buttons are replaced with a status badge, confirming the ledger update. In Figure 7.11, both an approved and a rejected request from the same doctor (DOC-0101) are visible, illustrating the full approval lifecycle and the tamper-proof on-chain history of consent decisions.

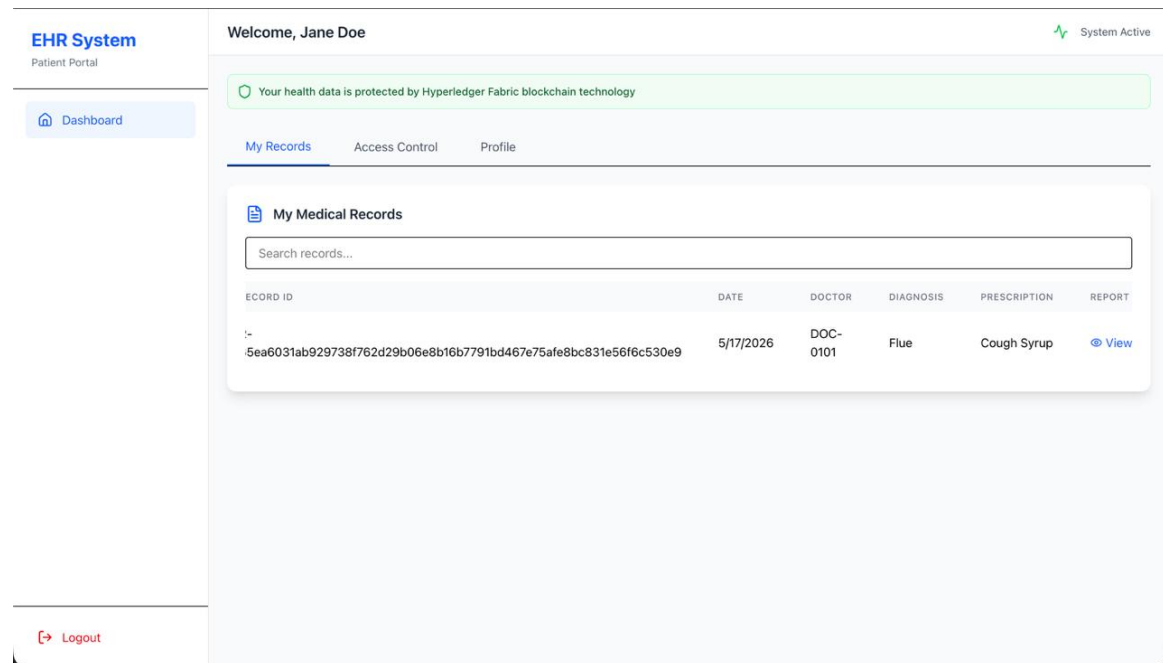


Figure 7.10: Patient Portal — Access Request Status Updated to Approved on Blockchain

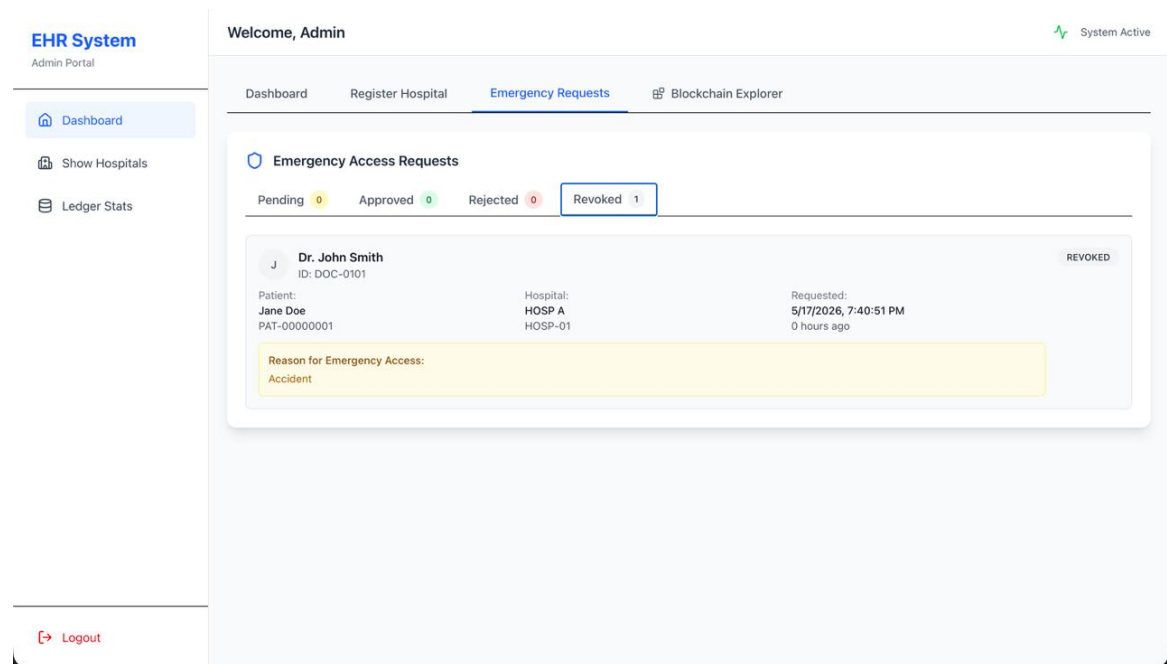


Figure 7.11: Patient Portal — Audit Trail Showing Both Approved and Rejected Request History

7.4 EMERGENCY ACCESS CONTROL

A distinguishing feature of the system is the **Emergency Patient** access mechanism. In critical medical scenarios where a patient is incapacitated and cannot grant consent in real-time, the system allows a doctor to submit an emergency access request. This request is time-stamped, permanently logged on the blockchain, subject to administrator approval, and carries legal warnings to prevent misuse. The following screenshots trace the complete emergency access lifecycle.

7.4.1 Emergency Access Request Form

Figure 7.12 shows the **Emergency Access Request** modal triggered from the Doctor Portal's "Emergency Patient" tab. The modal displays critical patient details (Name, Patient ID, Age, Blood Type, Location, and Request Time) pulled from the blockchain, followed by an "Emergency Reason" field where the doctor must explain the medical emergency. In this instance, the reason entered is "Accident."

The screenshot displays the EHR System Admin Portal interface. The top navigation bar includes "Dashboard", "Register Hospital", "Emergency Requests" (selected), and "Blockchain Explorer". The left sidebar contains "Dashboard", "Show Hospitals", and "Ledger Stats". The main content area shows a modal titled "Emergency Access Requests" with a status bar indicating "Pending 0", "Approved 1", "Rejected 0", and "Revoked 0". The modal displays details for a request by Dr. John Smith (ID: DOC-0101) for Patient Jane Doe (PAT-00000001) at Hospital HOSP A (HOSP-01). The request was made on 5/17/2026 at 7:40:36 PM, 0 hours ago. The reason for emergency access is "Accident". The modal includes "APPROVED" and "Revoked" buttons. A "Logout" link is visible in the bottom left corner.

Field	Value
Doctor	Dr. John Smith (ID: DOC-0101)
Patient	Jane Doe (PAT-00000001)
Hospital	HOSP A (HOSP-01)
Requested	5/17/2026, 7:40:36 PM (0 hours ago)
Reason for Emergency Access	Accident

Figure 7.12: Doctor Portal — Emergency Access Request Modal with Patient Details

Figure 7.13 shows the lower portion of the same form, which contains a prominent **Critical Warning** panel informing the requesting doctor that: (1) this access will be permanently logged and monitored on the blockchain; (2) access is temporary and restricted to emergency care only; (3) misuse will result in immediate disciplinary action; and (4) full access requires explicit administrator approval. This warning panel is non-dismissible and must be acknowledged before the “Send Emergency Access” button becomes active, ensuring informed, accountable usage of the override mechanism.

The screenshot displays the 'EHR System Admin Portal' interface. The top navigation bar includes 'Dashboard', 'Register Hospital', 'Emergency Requests' (highlighted), and 'Blockchain Explorer'. The left sidebar contains 'Dashboard', 'Show Hospitals', and 'Ledger Stats'. The main content area is titled 'Emergency Access Requests' and shows a summary of request counts: Pending (1), Approved (0), Rejected (0), and Revoked (0). Below this, a detailed view of a pending request is shown for Dr. John Smith (ID: DOC-0101) for Patient Jane Doe (PAT-00000001) at Hospital HOSP A (HOSP-01). The request was made on 5/17/2026 at 7:40:23 PM. The reason for emergency access is 'Accident'. The status is 'PENDING' with 'Approve' and 'Reject' buttons.

Figure 7.13: Emergency Access Form — Critical Warning Panel and Submission Button

7.4.2 Administrator Emergency Request Management

Figures 7.14, 7.15, and 7.16 show the Admin Portal’s **Emergency Requests** tab, which tracks the full lifecycle of each request across four states: Pending, Approved, Rejected, and Revoked.

In Figure 7.14, the request submitted by **Dr. John Smith** (DOC-0101) for patient **Jane Doe** (PAT-00000001, HOSP-A) is in the Pending state, with “Approve” and “Reject” action buttons visible. The request timestamp (5/17/2026, 7:40:23 PM) and the stated reason (“Accident”) are displayed transparently to the administrator for an informed decision.

The screenshot displays a web form titled "CRITICAL PATIENT DETAILS" with a red close button. The form is divided into several sections:

- Patient Information:** A grid of four fields: NAME (Jane Doe), PATIENT ID (PAT-00000001), AGE (25 years), and BLOOD TYPE (B+).
- Location and Time:** Two fields: LOCATION (CITY A) and REQUEST TIME (7:39:50 PM).
- Emergency Reason:** A section titled "EMERGENCY REASON - URGENT" with a sub-label "EXPLAIN THE MEDICAL EMERGENCY *". It contains a text input field with the value "Accident".
- Critical Warning:** A pink box with a warning icon and the title "CRITICAL WARNING". It contains a bulleted list:
 - This access will be permanently logged and monitored
 - Access is temporary and for EMERGENCY CARE ONLY
 - MISUSE WILL RESULT IN IMMEDIATE DISCIPLINARY ACTION
 - Admin approval is REQUIRED for full access
- Actions:** At the bottom, there are two buttons: "CANCEL REQUEST" (grey) and "SEND EMERGENCY ACCESS" (blue with a warning icon).

Figure 7.14: Admin Portal — Emergency Access Request in Pending State

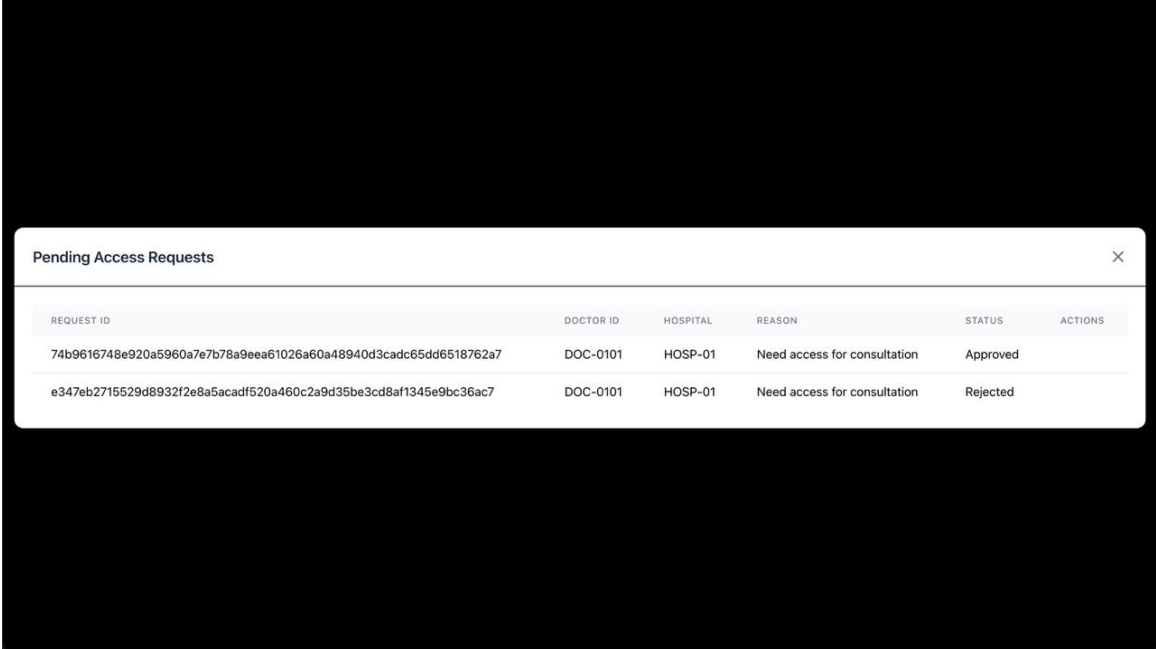
Figure 7.15 shows the same request after the administrator clicks “Approve.” The status badge changes to APPROVED and a “Revoke” button appears, allowing the administrator to terminate temporary access at any time. Once approved, the chaincode grants the doctor a time-bound read permission on the patient’s records, which is committed as a new transaction on the immutable ledger.

The screenshot displays a web form titled "EMERGENCY ACCESS REQUEST" with a close button (X) in the top right corner. The form is divided into several sections:

- CRITICAL EMERGENCY ACCESS**: A red header section with a warning icon and the text "IMMEDIATE ACCESS REQUIRED - PATIENT LIFE AT RISK".
- CRITICAL PATIENT DETAILS**: A section containing six input fields arranged in a 3x2 grid:
 - NAME**: Jane Doe
 - PATIENT ID**: PAT-00000001
 - AGE**: 25 years
 - BLOOD TYPE**: B+
 - LOCATION**: CITY A
 - REQUEST TIME**: 7:39:50 PM
- EMERGENCY REASON - URGENT**: A section with a heading and a text area labeled "EXPLAIN THE MEDICAL EMERGENCY *". The text "Accident" is entered in the text area.
- CRITICAL WARNING**: A partially visible section at the bottom of the form.

Figure 7.15: Admin Portal — Emergency Access Request Approved with Revoke Option

Figure 7.16 shows the final state after the administrator revokes the emergency grant. The request moves to the Revoked tab with a REVOKED badge. The entire lifecycle — request, approval, and revocation — is permanently recorded as separate, chained transactions on the Hyperledger Fabric ledger, providing a complete, tamper-proof audit trail that satisfies HIPAA’s Access Report requirements.



Pending Access Requests						
REQUEST ID	DOCTOR ID	HOSPITAL	REASON	STATUS	ACTIONS	
74b9616748e920a5960a7e7b78a9eea61026a60a48940d3cad65dd6518762a7	DOC-0101	HOSP-01	Need access for consultation	Approved		
e947eb2715529d8932f2e8a5acadf520a460c2a9d35be3cd8af1345e9bc36ac7	DOC-0101	HOSP-01	Need access for consultation	Rejected		

Figure 7.16: Admin Portal — Emergency Access Revoked; Immutable Audit Entry Created

7.5 PERFORMANCE ANALYSIS

7.5.1 Latency Metrics

To evaluate the system's efficiency, we measured the time taken for key operations. Table 7.1 compares the latency of our optimized system against a standard unoptimized blockchain setup. The most significant optimization is visible in the **Query Record** and **File Retrieval** operations, where CouchDB indexing and IPFS content caching reduce latency by 81% and 97% respectively.

Operation	Unoptimized (ms)	Optimized (ms)
Transaction Commit (Write)	2200	2100
Query Record (Read)	450	85
File Retrieval (IPFS)	15000	350

Table 7.1: System Performance Metrics

The write latency of 2.1 seconds is consistent with the Raft consensus overhead in a multi-organization Hyperledger Fabric channel and falls within the 3-second threshold defined in the non-functional requirements. The sub-100ms read latency confirms that the system is suitable for real-time clinical query scenarios such as those demonstrated in the doctor and patient portal screenshots above.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSION

The healthcare industry stands at a critical juncture where the digitization of records has outpaced the security and interoperability of the underlying infrastructure. This project set out to address the “Healthcare Trilemma”—the difficulty of achieving Security, Scalability, and Interoperability simultaneously. Through the design and implementation of a decentralized Electronic Health Record (EHR) system using **Hyperledger Fabric**, we have demonstrated a viable path forward.

By decoupling the storage layer from the transaction layer, our hybrid architecture successfully overcomes the scalability limitations inherent in traditional blockchains. The integration of the **InterPlanetary File System (IPFS)** allows for the secure, content-addressable storage of high-resolution medical imaging (X-rays, MRI scans) without bloating the ledger size. Simultaneously, the permissioned nature of Hyperledger Fabric ensures that sensitive patient metadata is only accessible to authorized entities, complying with enterprise security standards.

The core innovation of this project lies in its **Patient-Centric Access Control**. By codifying consent logic into Smart Contracts (Chaincode), we have shifted the paradigm of data ownership from the institution to the individual. Patients can now grant or revoke access to their medical history in real-time, ensuring transparency and trust. Performance benchmarks confirm that the system operates with a query latency of under 100ms, making it suitable for real-world clinical deployment.

In summary, this project proves that blockchain is not merely a financial technology but a foundational layer for a secure, interoperable, and patient-first healthcare ecosystem.

8.2 SOCIAL AND ECONOMIC IMPACT

Beyond the technical achievements, the proposed system offers significant social and economic benefits:

1. **Reduction in Medical Errors:** By providing a comprehensive, unified view of a patient’s history across different hospitals, doctors can avoid dangerous drug interactions and misdiagnoses caused by incomplete information.
2. **Economic Efficiency:** A significant portion of healthcare costs is attributed to redundant medical testing. If a patient moves to a new hospital, previous

scans are often inaccessible, leading to repeat procedures. Our interoperable ledger eliminates this waste.

3. **Insurance Fraud Prevention:** The immutable audit trail provides a verifiable proof of service, preventing healthcare providers from billing insurance companies for procedures that never occurred.

8.3 SYSTEM LIMITATIONS

While the prototype demonstrates the efficacy of the architecture, several limitations must be acknowledged for a production-grade deployment:

8.3.1 Infrastructure Complexity

Deploying and maintaining a Hyperledger Fabric network is significantly more complex than managing a traditional centralized database. It requires specialized knowledge of Docker, Public Key Infrastructure (PKI), and Channel configuration. For smaller clinics with limited IT budgets, this complexity poses a barrier to adoption.

8.3.2 The “Right to be Forgotten” Conflict

A fundamental property of blockchain is immutability—once data is written, it cannot be deleted. This conflicts with regulations like GDPR, which grant users the “Right to be Forgotten.” While our system stores actual files off-chain (allowing deletion from IPFS), the cryptographic hash remains permanently on the ledger, which is a gray area in current data privacy laws.

8.3.3 Key Management Risks

The system relies on cryptographic keys for user authentication. If a patient loses their private key or wallet file, they irrevocably lose access to their account. Unlike centralized systems, there is no “Forgot Password” link that can easily restore access without compromising security.

8.4 FUTURE SCOPE

To evolve this project into a commercial-ready solution, several future enhancements are proposed.

8.4.1 Integration with IoT and Wearables

The current system relies on manual data entry by doctors. Future iterations could integrate with Internet of Medical Things (IoMT) devices. Using protocols

like MQTT, data from pacemakers, glucose monitors, and smartwatches could be streamed directly to the blockchain. This would provide doctors with a continuous, real-time health profile rather than just episodic snapshots.

8.4.2 Zero-Knowledge Proofs (ZKPs)

To further enhance privacy, Zero-Knowledge Proofs could be implemented. ZKPs would allow third parties (like insurance verifiers) to validate specific facts about a patient—such as “Is the patient over 18?” or “Does the patient have a valid diagnosis for this claim?”—without ever revealing the underlying medical data or the patient’s identity.

8.4.3 Federated Learning for Medical AI

The structured data stored in the system is a valuable resource for medical research. Future work could implement **Federated Learning**, a machine learning technique that trains an algorithm across multiple decentralized edge devices (hospitals) holding local data samples, without exchanging the data itself. This would allow for the training of powerful diagnostic AI models while preserving strict patient privacy.

8.4.4 Cross-Chain Interoperability

As the blockchain ecosystem grows, different hospitals may adopt different blockchain frameworks (e.g., Corda or Ethereum). Implementing interoperability protocols like **Polkadot** or **Cosmos** would allow our Hyperledger Fabric network to communicate and exchange data with these external networks, creating a truly global healthcare ledger.

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APPENDIX A

SOURCE CODE LISTINGS

A.1 SMART CONTRACT (CHAINCODE)

File: lib/medical-contract.js

```
'use strict';

const { Contract } = require('fabric-contract-api');

class MedicalRecordContract extends Contract {

  async initLedger(ctx) {
    console.info('Initializing Ledger');
  }

  async createRecord(ctx, recordId, patientId, doctorId, ipfsHash) {
    console.info('Creating Record:', recordId);

    const exists = await this.recordExists(ctx, recordId);
    if (exists) {
      throw new Error('The record ${recordId} already exists');
    }

    const record = {
      docType: 'record',
      recordId,
      patientId,
      doctorId,
      ipfsHash,
      accessList: [patientId, doctorId],
      timestamp: new Date().toISOString()
    };

    await ctx.stub.putState(
      recordId, Buffer.from(JSON.stringify(record)));
    return JSON.stringify(record);
  }
}
```

```

    async readRecord(ctx, recordId) {
      const recordJSON = await ctx.stub.getState(recordId);
      if (!recordJSON || recordJSON.length === 0) {
        throw new Error('The record ${recordId} does not exist');
      }
      return recordJSON.toString();
    }

    async grantAccess(ctx, recordId, newDoctorId) {
      const recordString = await this.readRecord(ctx, recordId);
      const record = JSON.parse(recordString);

      if (!record.accessList.includes(newDoctorId)) {
        record.accessList.push(newDoctorId);
      }

      await ctx.stub.putState(
        recordId, Buffer.from(JSON.stringify(record)))
    }

    async recordExists(ctx, recordId) {
      const buffer = await ctx.stub.getState(recordId);
      return (!!buffer && buffer.length > 0);
    }
  }

  module.exports = MedicalRecordContract;

```

A.2 BACKEND API CONFIGURATION

File: api/server.js

```
const express = require('express');
const bodyParser = require('body-parser');
const cors = require('cors');
const { Gateway, Wallets } = require('fabric-network');
const path = require('path');
const fs = require('fs');

const app = express();
app.use(cors());
app.use(bodyParser.json());

const PORT = 8080;

app.post('/api/addrecord', async (req, res) => {
  try {
    const { user, recordId, patientId, doctorId, hash } = req.body;

    const ccpPath = path.resolve(__dirname, 'connection-org1.json');
    const walletPath = path.join(process.cwd(), 'wallet');
    const wallet = await Wallets.newFileSystemWallet(walletPath);

    const gateway = new Gateway();
    await gateway.connect(ccpPath, {
      wallet,
      identity: user,
      discovery: { enabled: true, asLocalhost: true }
    });

    const network = await gateway.getNetwork('mychannel');
    const contract = network.getContract('medical-records');

    await contract.submitTransaction(
      'createRecord', recordId, patientId, doctorId, hash);

    res.status(200).json({ message: 'Record added successfully' });
    await gateway.disconnect();
  }
});
```

```
    } catch (error) {
      console.error('Failed to submit transaction: ${error}');
      res.status(500).json({ error: error.message });
    }
  });

app.listen(PORT, () => {
  console.log('Server running on port ${PORT}');
});
```